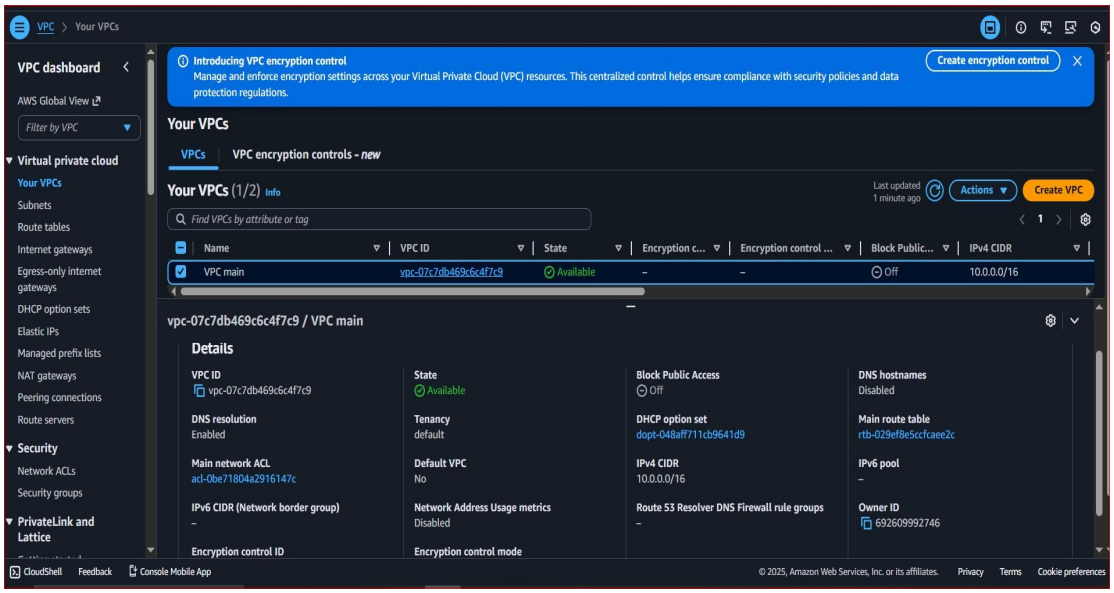
AWS Networking – Public EC2 Internet Access

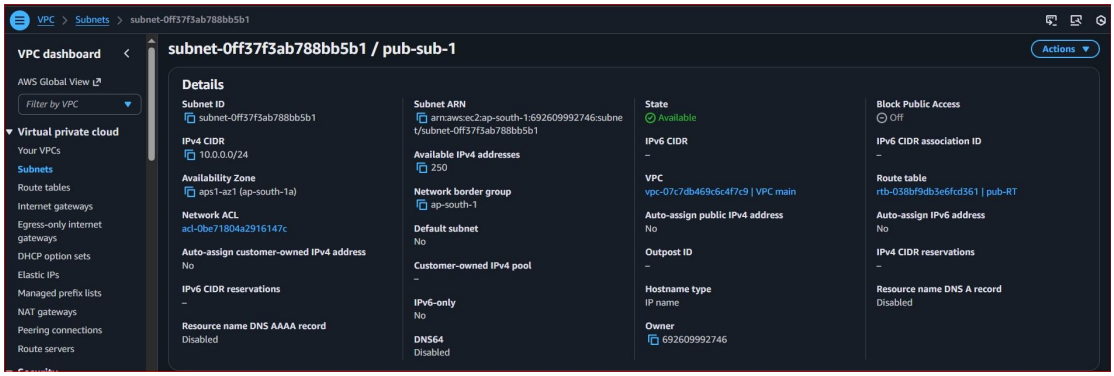
* Implemented secure internet connectivity using VPC subnets, route tables, Internet Gateway, and NAT Gateway.

Step 1 : Creating VPC

An Amazon Virtual Private Cloud (VPC) is a logically isolated virtual network in AWS that allows you to launch and manage AWS resources in a secure and controlled environment. It gives full control over networking components such as IP addressing, subnets, routing, and security.



Step 2 : Creating Subnet

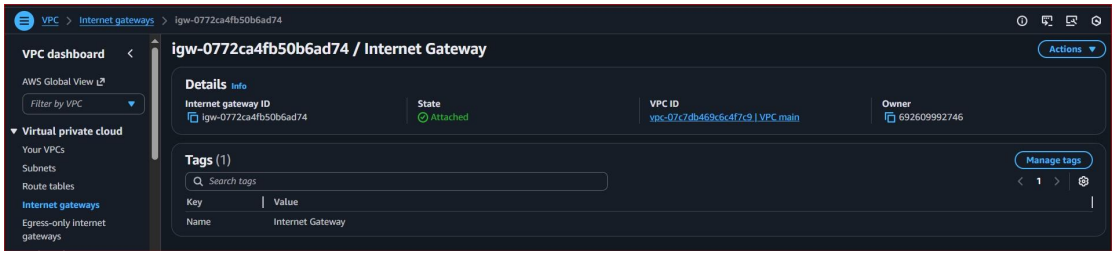


Step 3 : Creating Internet Gateway And Attaching VPC

An Internet Gateway (IGW) is a horizontally scaled, highly available AWS-managed component that enables communication between resources inside a VPC and the public internet.

The Internet Gateway is used to:

- Allow inbound and outbound internet traffic
- Enable public-facing resources to be accessible from the internet
- Provide a target for routes that send traffic outside the VPC

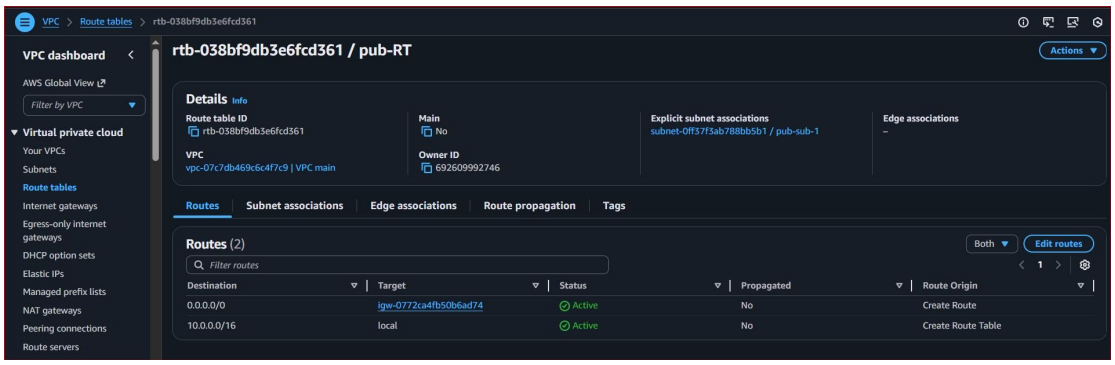


Step 4 : Creating Route Table (Public)

A Route Table in AWS is a set of rules (routes) that determines how network traffic is directed within a VPC. Each subnet must be associated with a route table, which controls where inbound and outbound traffic is sent.

- Route tables define how traffic moves inside and outside the VPC
- Public route tables use Internet Gateway
- Private route tables use NAT Gateway
- Proper routing ensures security and connectivity

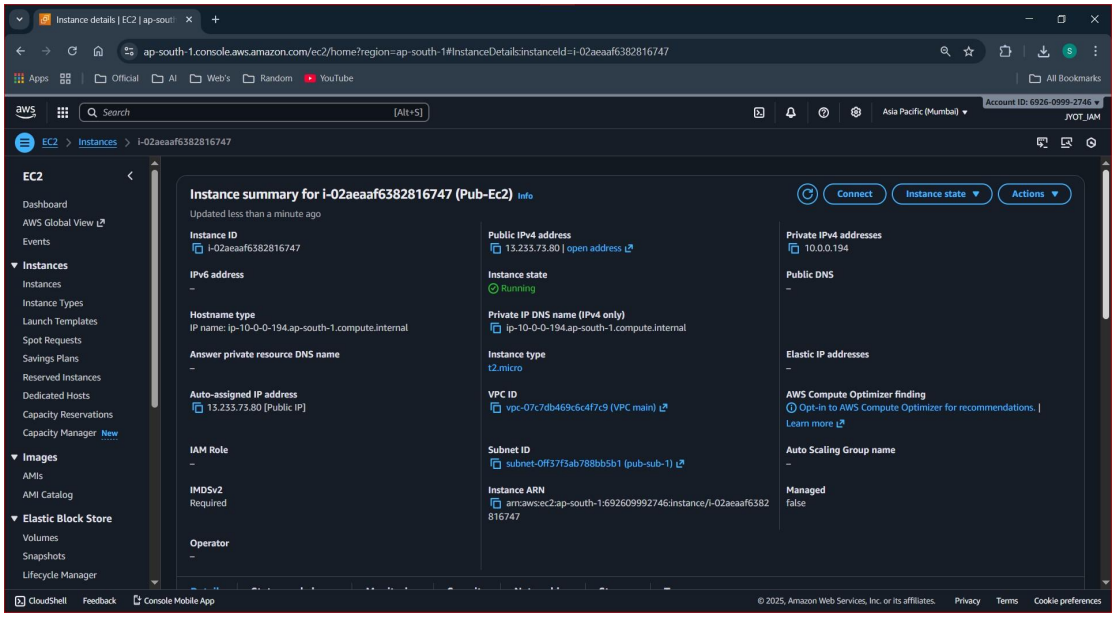
Note :- Routing from internet gateway to public route table
Routing from Public route table to public subnet



Step 5 : Creating Instance Using Networking Configuration

Launching an Amazon EC2 instance with the correct networking configuration is critical to ensure proper connectivity, security, and access control. In this architecture, EC2 instances are launched in public and private subnets within a VPC, based on their access requirements.

Note : Always enable HTTP/ HTTPS protocol in Security Group.



Step 6 : Verifying Internet Connectivity on EC2 Instance

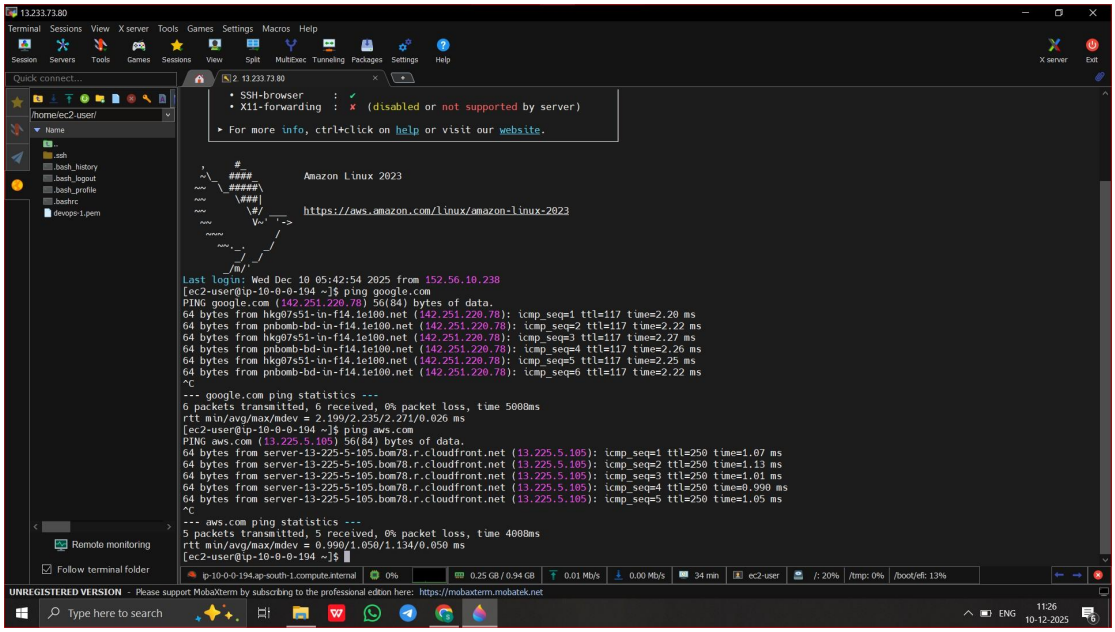
To verify that the EC2 instance has active internet connectivity after configuring networking components such as VPC, subnet, route table, Internet Gateway / NAT Gateway.

Internet connectivity was verified by testing communication with an external public domain.

ping google.com

Result

-The instance successfully resolved and reached google.com
-This confirms that the EC2 instance has proper internet connectivity
-Indicates correct configuration of:
-Route tables
-Internet Gateway (for public EC2) or NAT Gateway (for private EC2)
-Security Group outbound rules



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